```
# TASK 1: Loading audio and playing
import librosa as lb

y,sr=lb.load("D:\\Internship\\Day32\\game_over.wav", sr=None)
print("Audio duration: ",lb.get_duration(y=y, sr=sr),"seconds")
print("Sampling Rate:", sr,"Hz")

import sounddevice as sd
sd.play(y, sr)
sd.wait()

# TASK 2: Display waveform
import matplotlib.pyplot as plt
import librosa.display

plt.figure(figsize=(10,4))
lb.display.waveshow(y,sr=sr)

plt.title("Waveform")
plt.xlabel("Time")
plt.ylabel("Amplitude")
plt.show()

# TASK 3: Spectogram
import numpy as np

d=lb.amplitude_to_db(np.abs(lb.stft(y)),ref=np.max)
plt.figure(figsize=(10,4))
lb.display.specshow(d,sr=sr,x_axis="time",y_axis="log")
plt.title("Spectrogram")
plt.show()
```

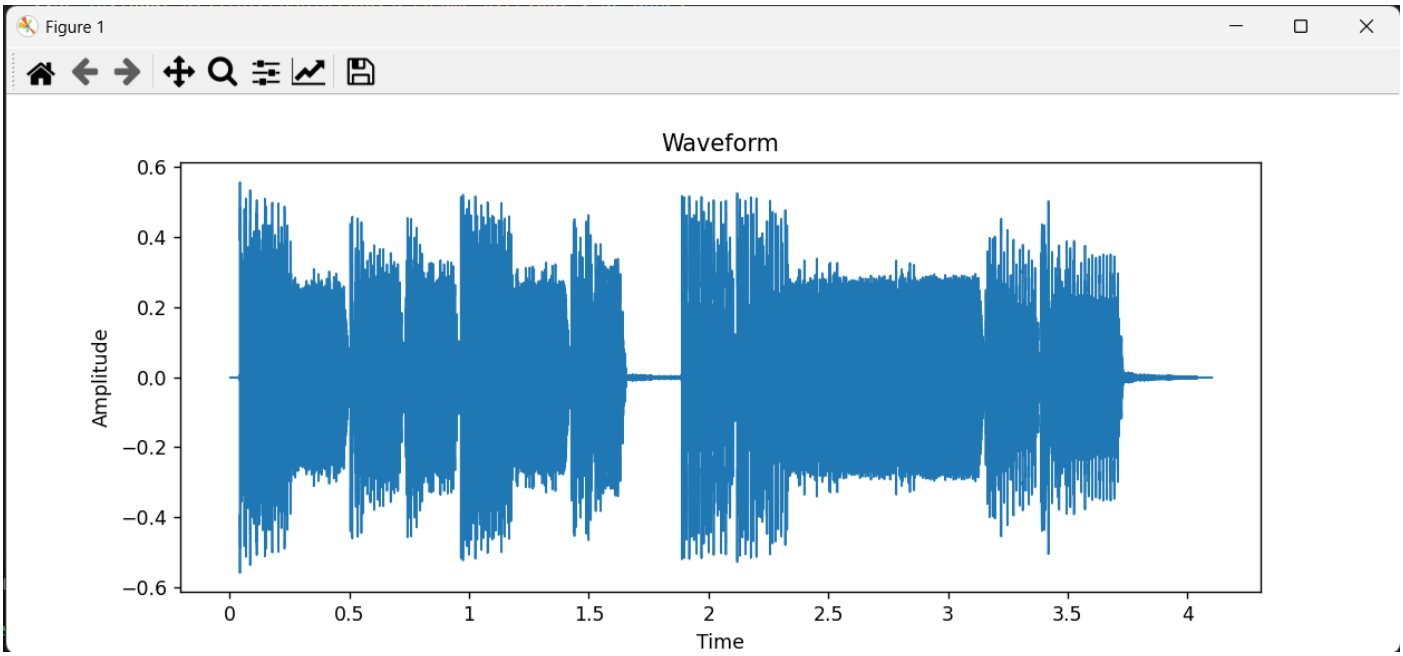
```
# TASK 4: Mel frequency cepstral coefficients
mfccs=lb.feature.mfcc(y=y,sr=sr,n_mfcc=13)

plt.figure(figsize=(10,4))
lb.display.specshow(mfccs,sr=sr,x_axis="time")

plt.title("MFCCs")
plt.show()

# TASK 5: Chroma feature
chroma=lb.feature.chroma_stft(y=y,sr=sr)

plt.figure(figsize=(10,4))
lb.display.specshow(chroma,sr=sr,x_axis="time",y_axis="chroma")

plt.title("Chroma features")
plt.show()

# TASK 6: Tempo and beat
tempo, beats=lb.beat.beat_track(y=y,sr=sr)
print("Estimated tempo:",tempo,"BPM")

beat_times=lb.frames_to_time(beats,sr=sr)

plt.figure(figsize=(10,4))
lb.display.waveshow(y,sr=sr)

plt.vlines(beat_times,ymin=-1,ymax=1,color='r',linestyle='--',label="Beats")
plt.legend()
plt.title("Beat tracking")

plt.show()
```

 # TASK 7: Zero crossing

```
z=lb.feature.zero_crossing_rate(y)


plt.figure(figsize=(10,4))
plt.plot(z[0],label="Zero crossing rate")


plt.xlabel("frames")
plt.ylabel("rate")
plt.title("zero crossing rate")


plt.legend()
plt.show()
```

<u>Outputs:</u>

```
(.venv) PS D:\Internship\Day32> python task1-7.py
Audio duration:  4.101224489795919 seconds
Sampling Rate: 44100 Hz
```



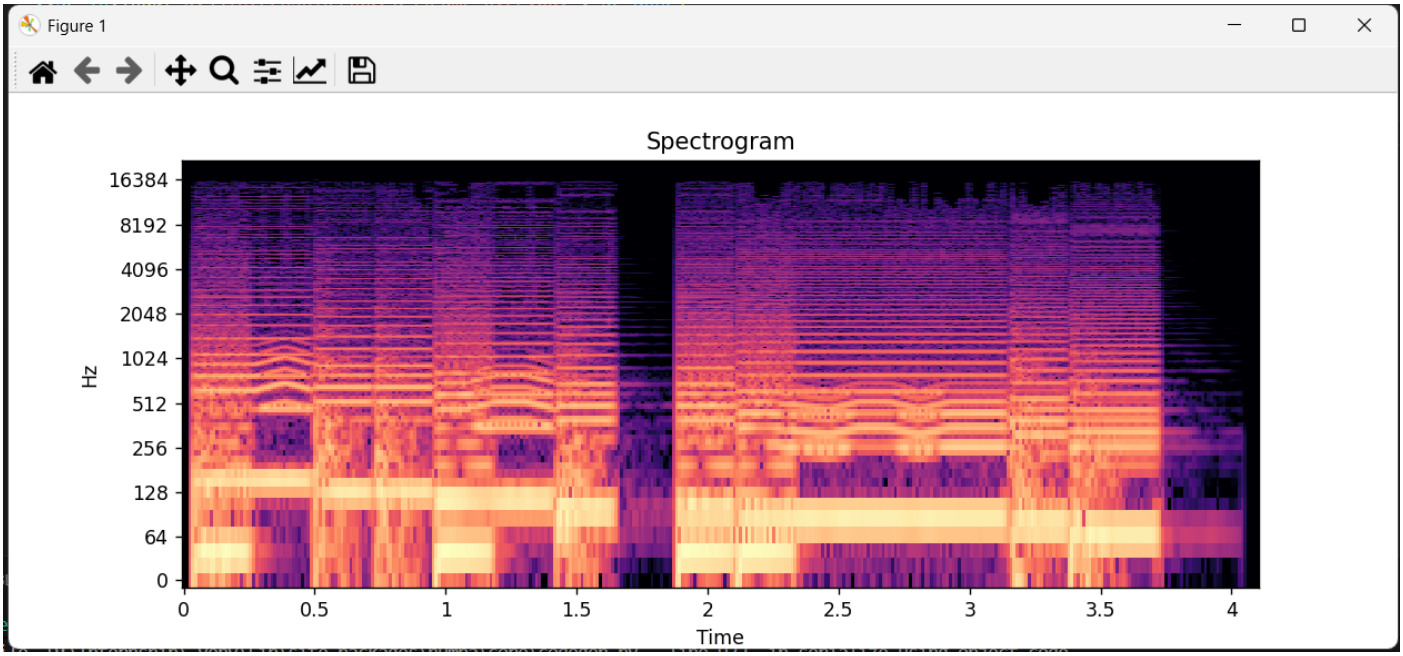


Figure 1
Spectrogram
16384
8192
4096
2048
1024
512
256
128
64
0
Hz
0
0.5
1
1.5
2
2.5
3
3.5
4
Time

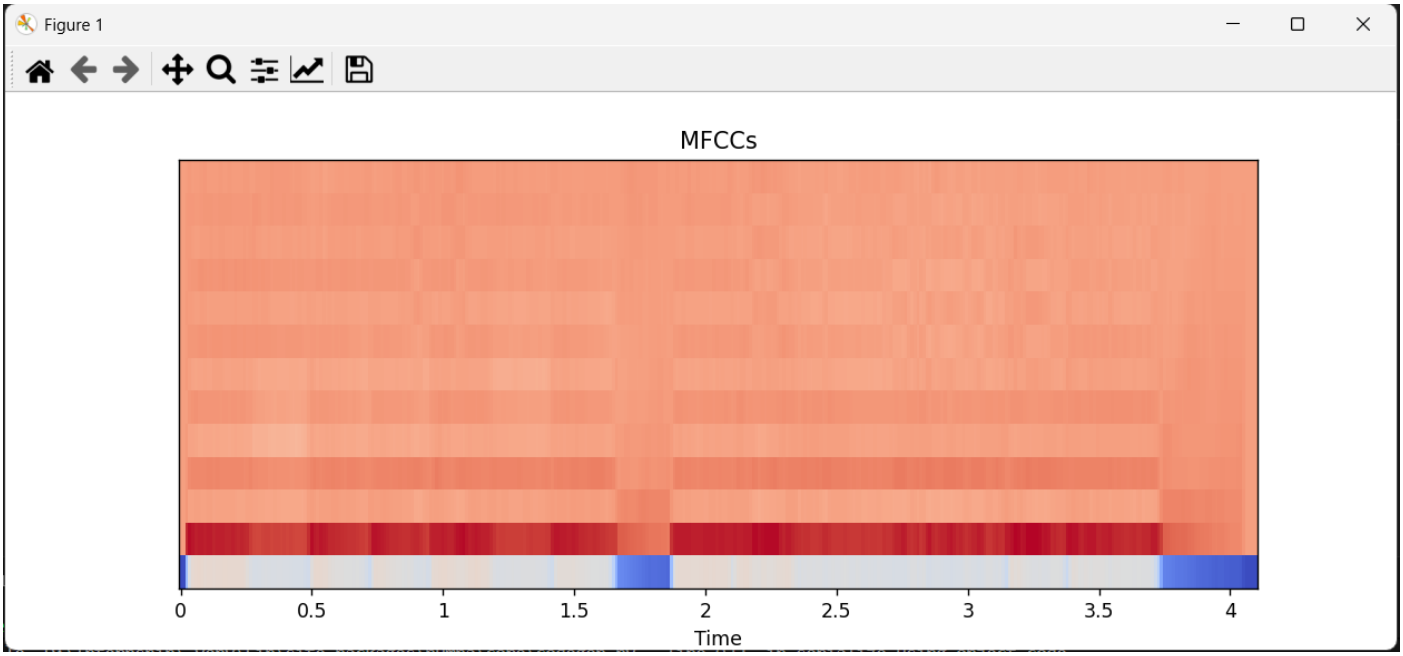


Figure 1
MFCCs
0
0.5
1
1.5
2
2.5
3
3.5
4
Time

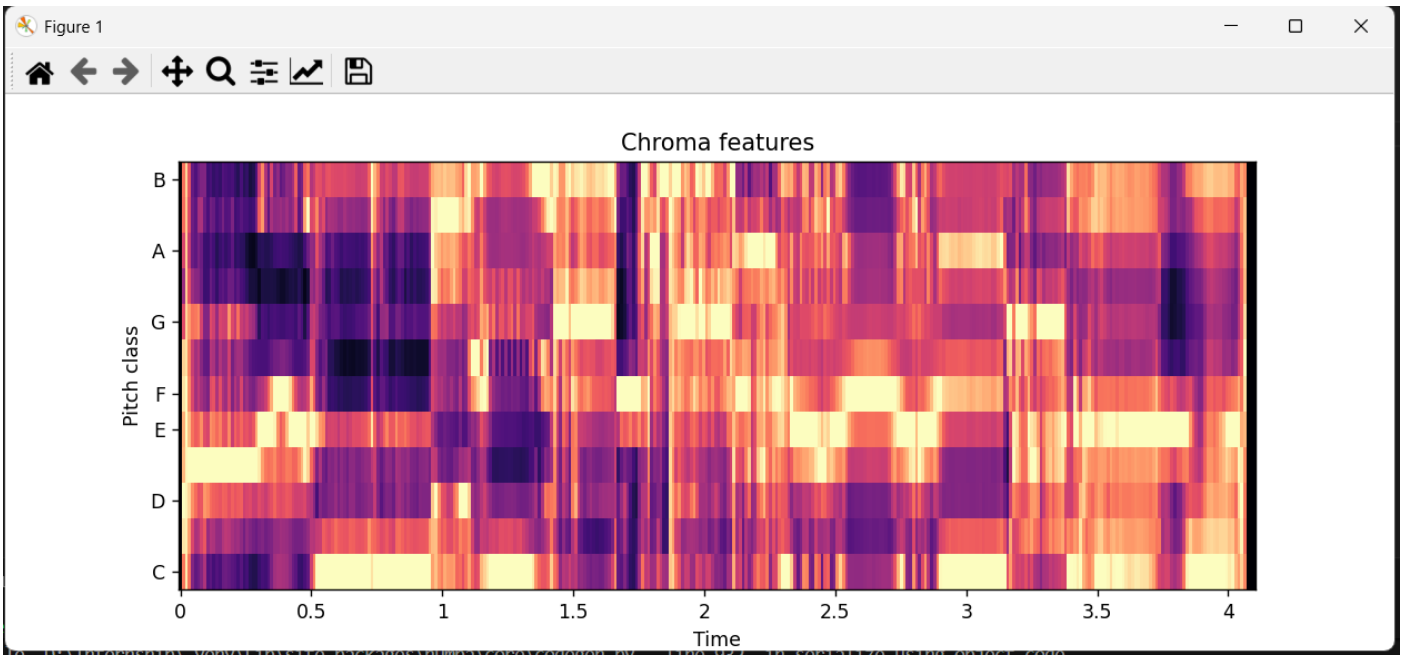


Figure 1
Chroma features
B
A
G
F
E
D
C
Pitch class
0
0.5
1
1.5
2
2.5
3
3.5
4
Time

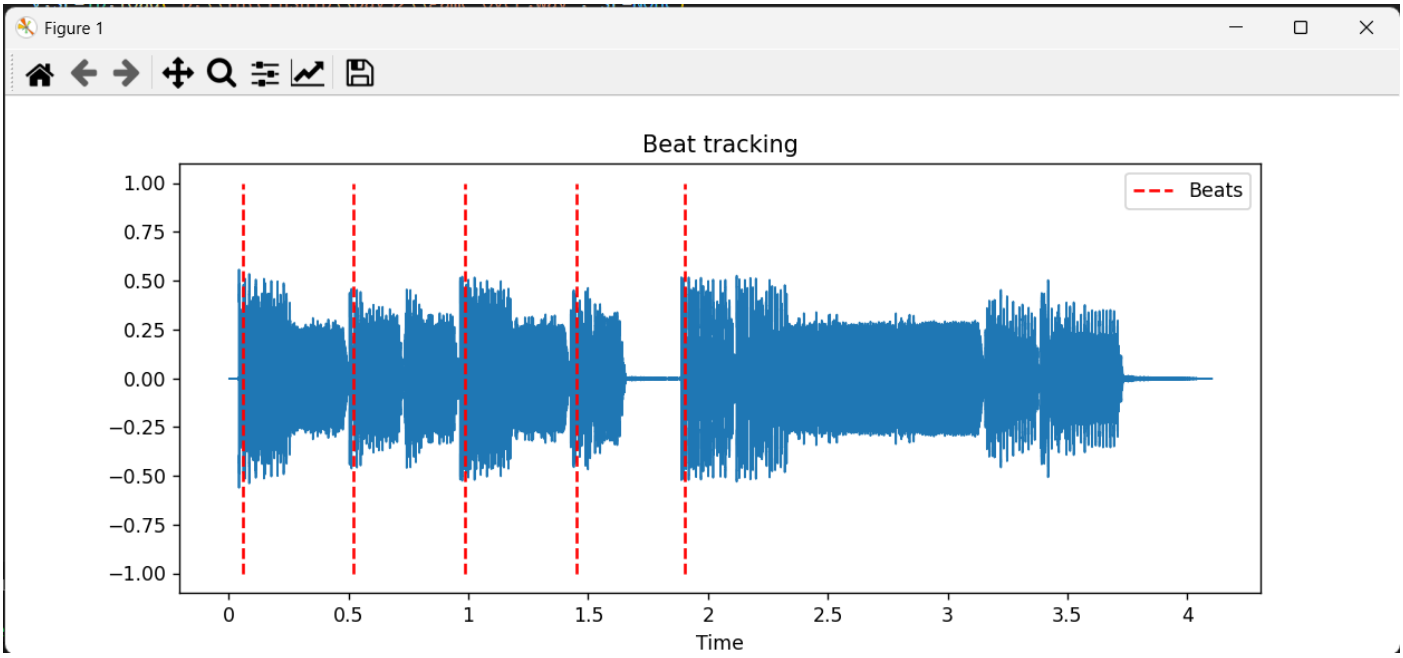


Figure 1
Beat tracking
Beats
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
0.5
1
1.5
2
2.5
3
3.5
4
Time

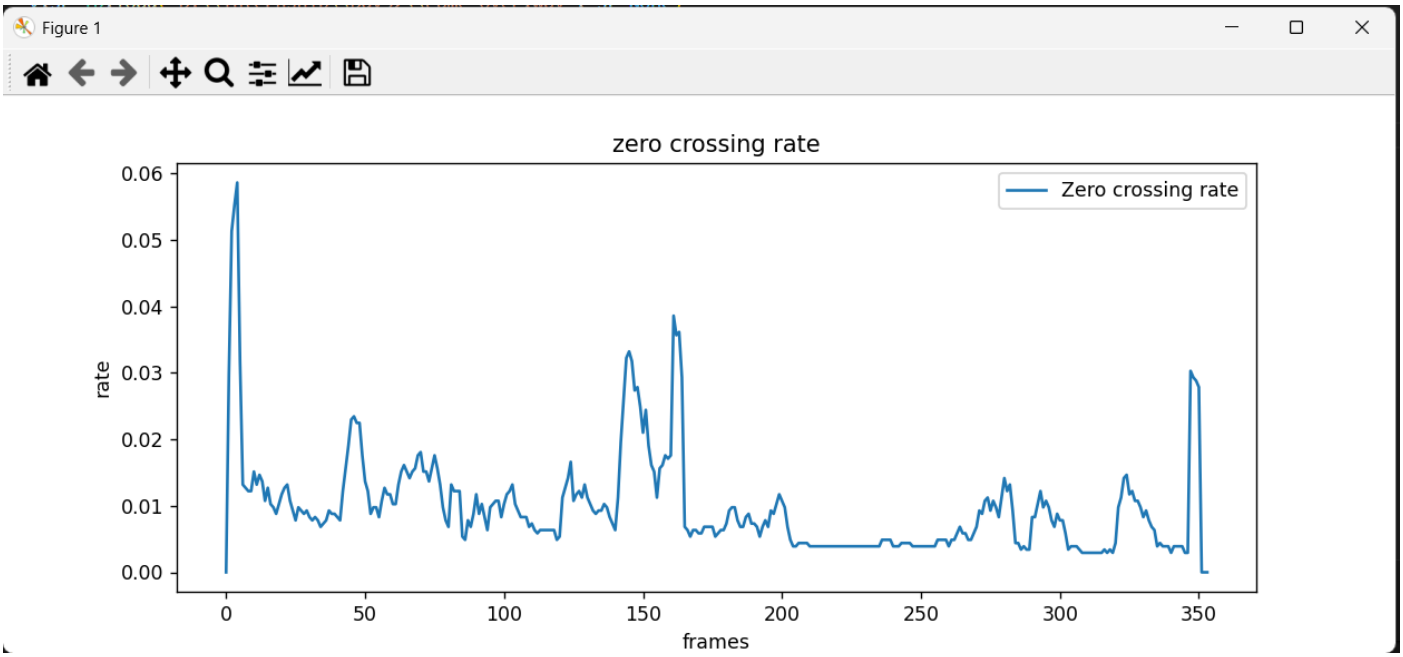


Figure 1
zero crossing rate
Zero crossing rate
0.06
0.05
0.04
0.03
0.02
0.01
0.00
rate
0
50
100
150
200
250
300
350
frames